

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO4	Assessment:	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	Identify the components and structure of mobile application for Android and Windows OS-based devices.		
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Section / Batch:	D / 3 rd Year	Date:	18.8.26

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 60 minutes
- Number of questions : 01

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

34. A company wants to develop a mobile application for an organization that has strict restrictions on third-party software and requires centralized application control. Compare iOS and Android in terms of security policies, application installation, sandboxing, permissions, and device management. Analyze which OS provides better administrative control.

1. Introduction:

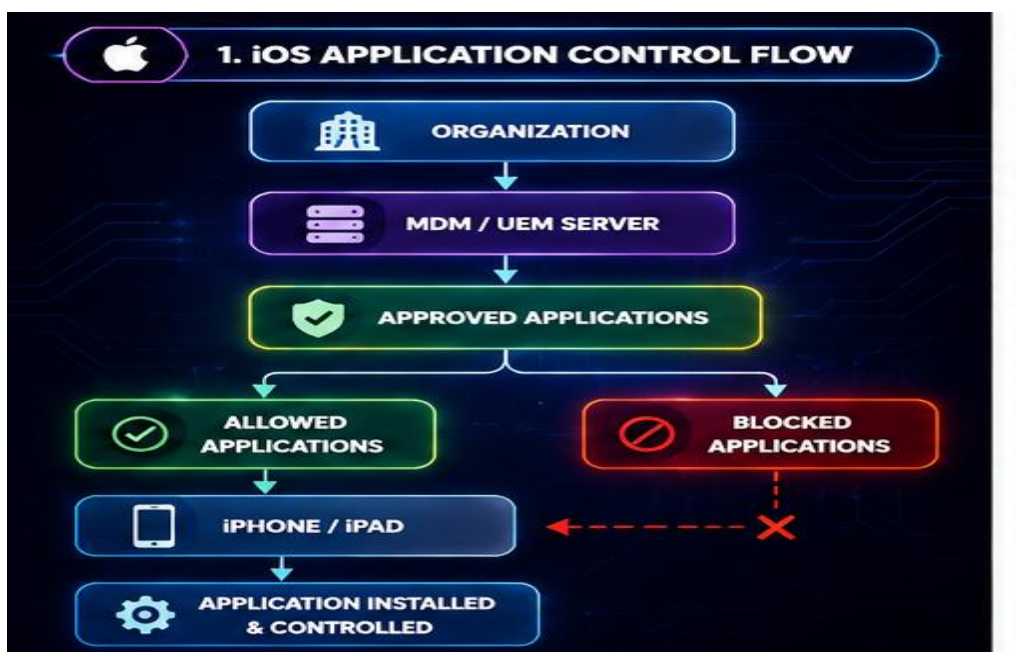
Mobile applications are now essential for communication, business operations, employee services, and access to enterprise information. For organizations handling confidential data, however, allowing employees complete freedom to install and configure applications can create security risks.

The two leading mobile operating systems, Apple iOS and Google Android, provide strong security mechanisms, but they follow different philosophies. iOS emphasizes a tightly controlled ecosystem, while Android emphasizes flexibility and customization.

This analysis compares both platforms in terms of security policies, application installation, sandboxing, permissions, and centralized device management. The objective is to identify which operating system provides better administrative control for the given organization.

2. Security Architecture at a Glance:

iOS uses secure boot, application signing, sandboxing, encryption, permission controls, and enterprise device management. Apple controls the hardware and software ecosystem, which helps maintain a consistent security environment.



3. Application Installation & Third-Party Software:

Application installation is a critical issue because the organization wants to prevent unauthorized third-party software. The administrator should be able to define an approved application list and prevent users from bypassing organizational policies.

iOS — Controlled Application Distribution

iOS primarily uses the App Store and provides enterprise distribution and device-management mechanisms. Through MDM/UEM, an organization can distribute approved applications, restrict application installation, and manage applications remotely.

Android — Flexible but Manageable

Android Enterprise and Managed Google Play allow administrators to distribute approved applications. Android can therefore support strict enterprise control, but its greater flexibility and device diversity mean that administrators must configure policies carefully to prevent unauthorized installation.



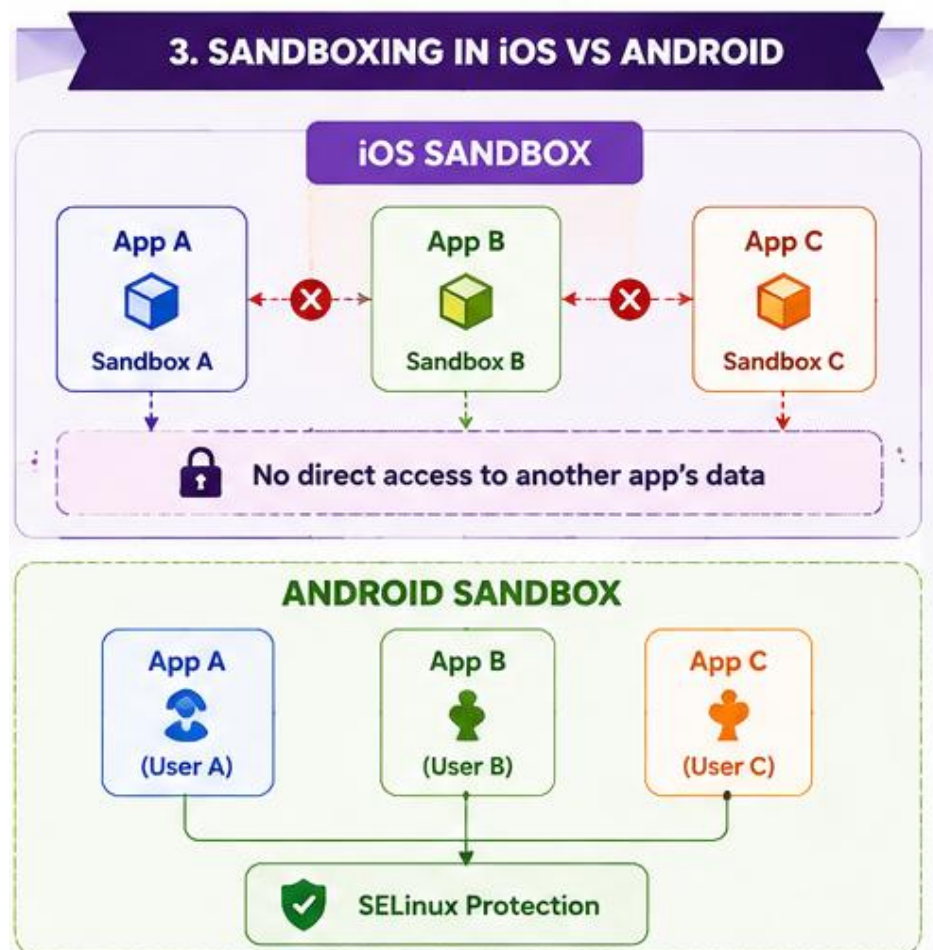
CONTROL AREA	iOS	ANDROID
Approved app distribution	Excellent	Excellent
Third-party restriction	Very strong	Strong with policy
Ecosystem consistency	Very high	Varies by device

4. Sandboxing & Permission Management:

Sandboxing isolates applications so that one application cannot freely access another application's private data. Both platforms use strong isolation, although their underlying implementations differ.

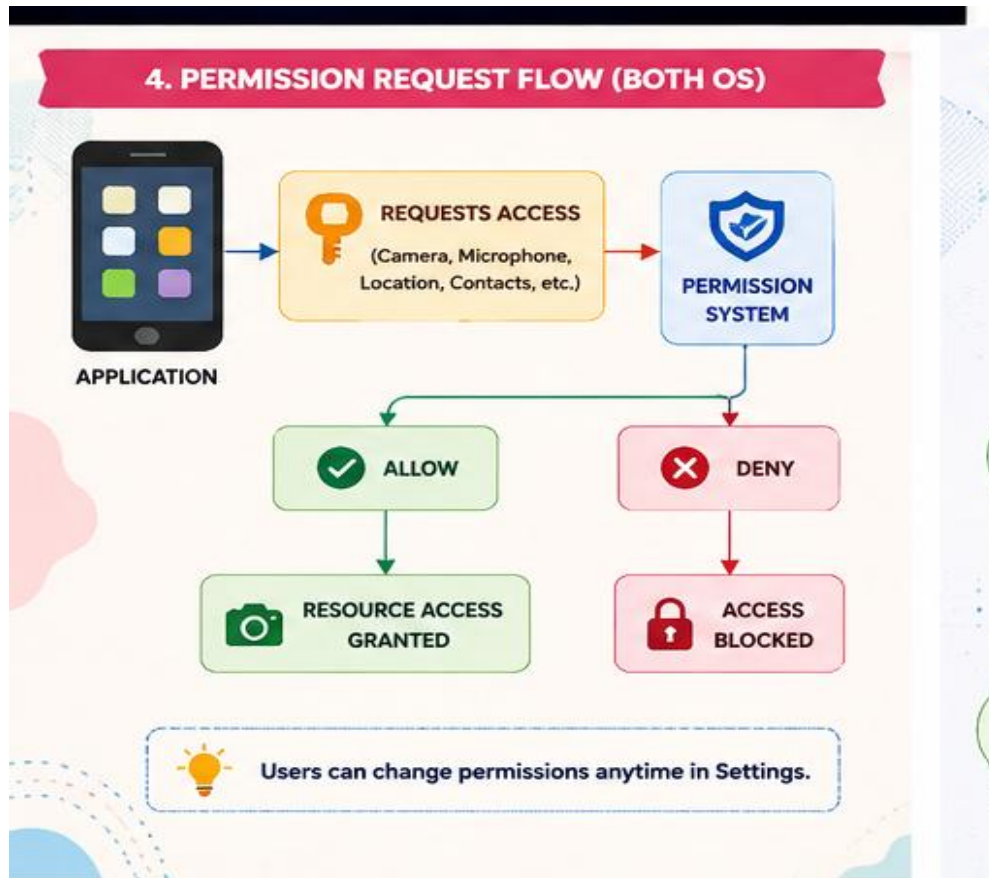
Sandboxing

On iOS, applications operate inside restricted sandboxes. Android also isolates applications using separate Linux identities and reinforces isolation with SELinux security policies. Both approaches reduce the impact of a compromised application.



🔑 Permission Management

Applications may request access to sensitive resources such as the camera, microphone, location, contacts, photos, and files. Both operating systems use permission systems that require access to be granted rather than automatically exposing protected resources.

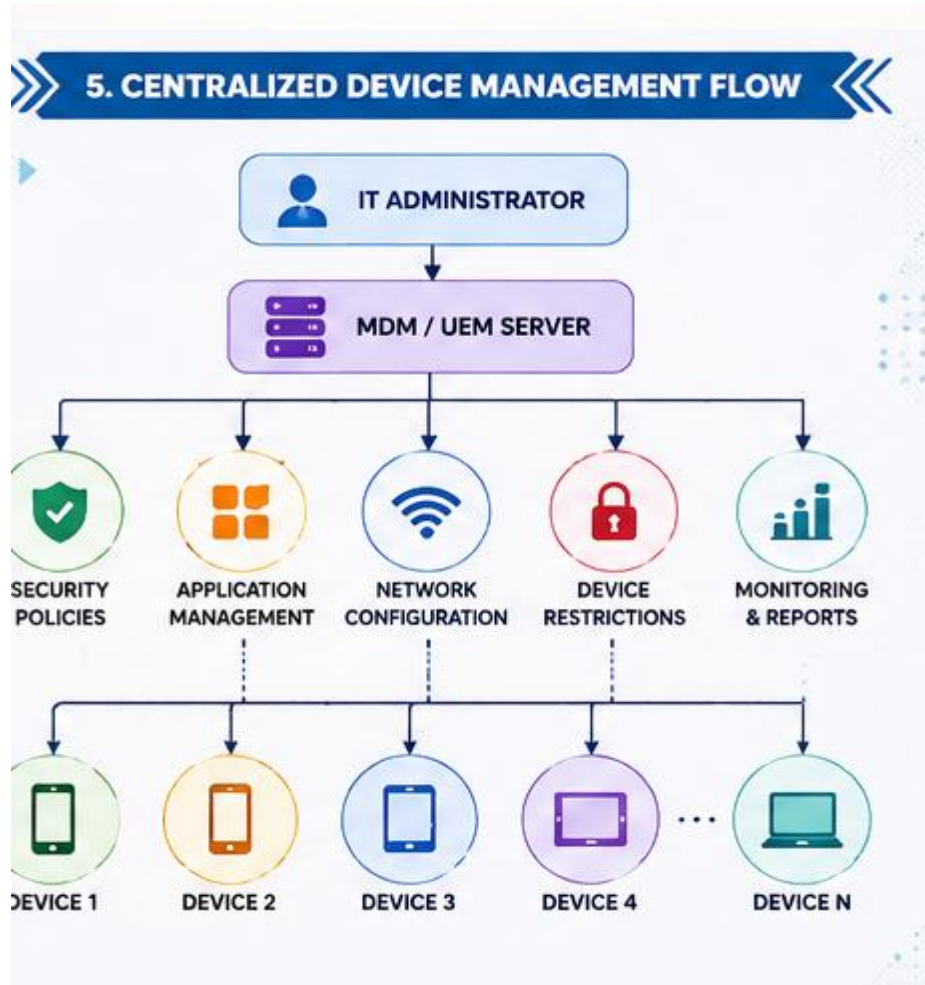


Permission Comparison :

FEATURE	iOS	ANDROID
Runtime permissions	✓	✓
Camera / microphone control	✓	✓
Location control	✓	✓
Enterprise permission management	Strong	Strong

5. Centralized Device Management

Centralized management is the most important requirement in the scenario. Mobile Device Management (MDM) or Unified Endpoint Management (UEM) allows the IT administrator to apply policies to many devices from a central control system.



Typical administrative functions include device enrollment, approved application distribution, password enforcement, network configuration, security policies, remote lock, remote wipe, application restrictions, and monitoring.

6. Head-to-Head Administrative Comparison:

REQUIREMENT	iOS	ANDROID	EDGE
Security policies	Excellent	Excellent	iOS
App installation control	Excellent	Very good	iOS
Sandboxing	Excellent	Excellent	Tie
Permissions	Excellent	Excellent	Tie

Centralized management	Excellent	Excellent	Tie / iOS consistency
Customization	Moderate	Excellent	Android
Device consistency	Very high	Varies	iOS

7. Final Recommendation:

For this specific organization, iOS provides the better overall administrative-control model. Android is also a powerful enterprise platform, but its flexibility and device diversity require careful configuration. iOS is more closely aligned with the stated need for strict third-party software restrictions and a consistent centrally managed environment.



FINAL VERDICT → 🏆 iOS is the better fit when centralized control, consistency, and strict application restrictions are the highest priorities.